

Loading Gold Layer AS Table

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/*
=====
DDL Script: Create Gold Tables
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Script Purpose:
    This script creates physical tables for the Gold layer in the data warehouse.
    The Gold layer represents the final dimension and fact tables (Star Schema).

    Each table is populated from the Silver layer after applying the required
    transformations and joins.

Usage:
    - These tables store the final business-ready data.
    - The tables can be queried directly for analytics and reporting.
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*/

-- =====
-- Create Dimension: gold.dim_customers
-- =====
IF OBJECT_ID('gold.dim_customers', 'U') IS NOT NULL
    DROP TABLE gold.dim_customers;
GO

CREATE TABLE gold.dim_customers (
    customer_key      INT,
    customer_id       INT,
    customer_number   NVARCHAR(50),
    first_name        NVARCHAR(50),
    last_name         NVARCHAR(50),
    country           NVARCHAR(50),
    marital_status    NVARCHAR(50),
    gender            NVARCHAR(50),
    birthdate         DATE,
    create_date       DATE
);
GO

INSERT INTO gold.dim_customers
SELECT
    ROW_NUMBER() OVER (ORDER BY cst_id) AS customer_key,
    ci.cst_id                          AS customer_id,
    ci.cst_key                         AS customer_number,
    ci.cst_firstname                   AS first_name,
    ci.cst_lastname                    AS last_name,
    la.cntry                          AS country,
    ci.cst_marital_status              AS marital_status,
    CASE
        WHEN ci.cst_gndr != 'n/a' THEN ci.cst_gndr
        ELSE COALESCE(ca.gen, 'n/a')
    END                               AS gender,
    ca.bdate                          AS birthdate,
    ci.cst_create_date                AS create_date
FROM silver.crm_cust_info ci
LEFT JOIN silver.erp_cust_az12 ca
    ON ci.cst_key = ca.cid
LEFT JOIN silver.erp_loc_a101 la
    ON ci.cst_key = la.cid;
```

GO

```
-- =====  
-- Create Dimension: gold.dim_products  
-- =====
```

```
IF OBJECT_ID('gold.dim_products', 'U') IS NOT NULL  
    DROP TABLE gold.dim_products;
```

GO

```
CREATE TABLE gold.dim_products (  
    product_key      INT,  
    product_id       INT,  
    product_number   NVARCHAR(50),  
    product_name     NVARCHAR(100),  
    category_id      NVARCHAR(50),  
    category         NVARCHAR(50),  
    subcategory      NVARCHAR(50),  
    maintenance      NVARCHAR(50),  
    cost             INT,  
    product_line     NVARCHAR(50),  
    start_date       DATE  
);
```

GO

```
INSERT INTO gold.dim_products  
SELECT
```

```
    ROW_NUMBER() OVER (ORDER BY pn.prd_start_dt, pn.prd_key) AS product_key,  
    pn.prd_id          AS product_id,  
    pn.prd_key         AS product_number,  
    pn.prd_nm          AS product_name,  
    pn.cat_id          AS category_id,  
    pc.cat             AS category,  
    pc.subcat          AS subcategory,  
    pc.maintenance     AS maintenance,  
    pn.prd_cost        AS cost,  
    pn.prd_line        AS product_line,  
    pn.prd_start_dt    AS start_date
```

```
FROM silver.crm_prd_info pn  
LEFT JOIN silver.erp_px_cat_glv2 pc  
    ON pn.cat_id = pc.id  
WHERE pn.prd_end_dt IS NULL;  
GO
```

```
-- =====  
-- Create Fact Table: gold.fact_sales  
-- =====
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```
IF OBJECT_ID('gold.fact_sales', 'U') IS NOT NULL  
    DROP TABLE gold.fact_sales;
```

GO

```
CREATE TABLE gold.fact_sales (  
    order_number     NVARCHAR(50),  
    product_key      INT,  
    customer_key     INT,  
    order_date       DATE,  
    shipping_date    DATE,  
    due_date         DATE,  
    sales_amount     INT,  
    quantity         INT,  
    price            INT  
);
```

GO

INSERT INTO gold.fact_sales

SELECT

sd.sls_ord_num AS order_number,
pr.product_key AS product_key,
cu.customer_key AS customer_key,
sd.sls_order_dt AS order_date,
sd.sls_ship_dt AS shipping_date,
sd.sls_due_dt AS due_date,
sd.sls_sales AS sales_amount,
sd.sls_quantity AS quantity,
sd.sls_price AS price

FROM silver.crm_sales_details sd

LEFT JOIN gold.dim_products pr

ON sd.sls_prd_key = pr.product_number

LEFT JOIN gold.dim_customers cu

ON sd.sls_cust_id = cu.customer_id;

GO